Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Kairos AI
b. Team leader name: Puneet Pahuja
c. Problem Statement: Manufacturing - Intelligent Operations & Industrial Efficiency: autonomous remote fault diagnosis & self-healing for power devices to cut downtime

Team members: Khushboo Verma, Anurag Verma

Brief about the idea

Kairos AI puts an expert reliability engineer inside every power device.

It learns each device from its own documentation - datasheets, register maps and event-log structure - using retrieval-augmented generation on Vertex AI, then continuously monitors live telemetry and event logs remotely.

A five-agent engine detects anomalies, diagnoses the root cause with cited evidence, decides with confidence-gated autonomy, acts to self-heal or escalate (human-in-the-loop), and verifies the fix - quantifying the downtime and cost avoided. The same engine serves both historical and live analysis.

Built end-to-end on Google Cloud: Gemini 2.5 Flash on Vertex AI, Vertex embeddings, Firestore, Cloud Storage, BigQuery and Secret Manager - deployed on Cloud Run.

Opportunities

How different is it from existing ideas?

Existing tools only alert. Kairos AI explains the root cause, correlates cascading alarms across the dependency chain, and closes the loop - decide, act, verify - with human approval, not just detection.

How will it solve the problem?

Remote, autonomous diagnosis in seconds removes the wait for an on-site engineer, cutting downtime. Every answer is grounded in the device's own manuals, so a junior technician gets senior-grade insight.

USP of the proposed solution

Self-configuring monitoring - the agent reads the datasheet to pick parameters, thresholds and decode event logs - plus one engine for live and historic data, and honest, cited, human-in-the-loop AI on a deep Google Cloud stack.

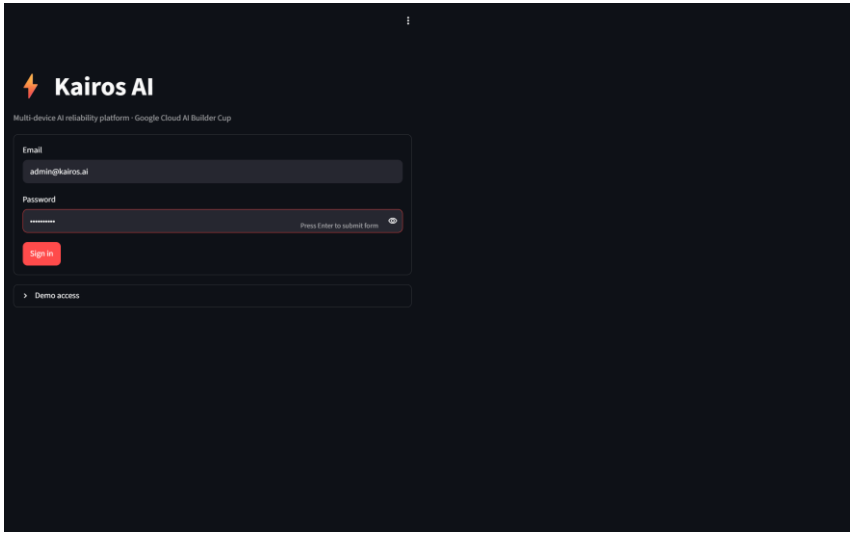
List of features offered by the solution

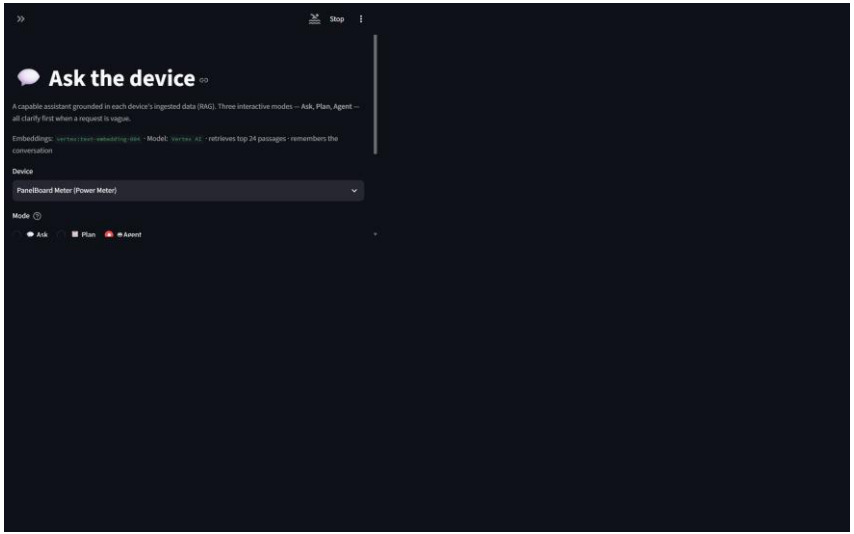
- Role-based multi-device platform (admin / engineer / viewer) with hashed authentication.
- Per-device knowledge base via RAG using Vertex AI embeddings (text-embedding-004).
- Knowledge ingestion: upload PDF / Word / Excel, or pull from GitHub and Confluence.
- Ask the device - agentic assistant with Ask / Plan / Agent modes; generates datasets, test cases and code.
- Five-agent analysis engine: Monitor -> Diagnose -> Decide -> Act -> Verify.
- Datasheet-driven live monitoring over REST, with automatic event-log decoding.
- Quantifies diagnosis accuracy, downtime avoided and cost avoided against a baseline.
- One engine for both historical datasets and live streams; secure secrets via Secret Manager.

Process flow diagram or Use-case diagram

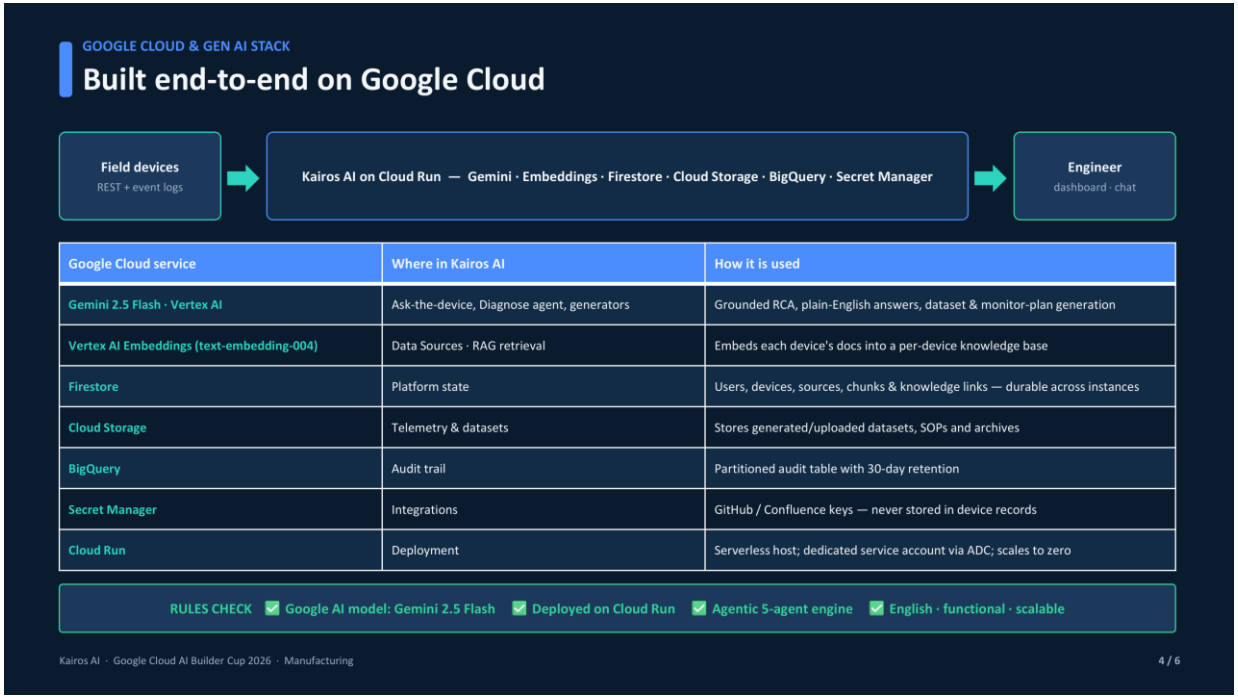


Wireframes/Mock diagrams of the proposed solution (optional)





Architecture diagram of the proposed solution



Technologies to be used in the solution

- Gemini 2.5 Flash on Vertex AI - grounded reasoning, root-cause analysis and generation.
- Vertex AI Embeddings (text-embedding-004) - RAG retrieval over device documents.
- Firestore - platform state: users, devices, sources, chunks and knowledge links.
- Cloud Storage - telemetry and generated/uploaded datasets.
- BigQuery - partitioned audit trail with 30-day retention.
- Secret Manager - integration keys, never stored in device records.
- Cloud Run - serverless deployment; dedicated service account via ADC; scales to zero.
- Python, Streamlit, scikit-learn (robust-Z + Isolation Forest) for the engine and UI.

Rules check: uses a Google AI model (Gemini) and is deployed on Google Cloud via Cloud Run.

Estimated implementation cost (optional)

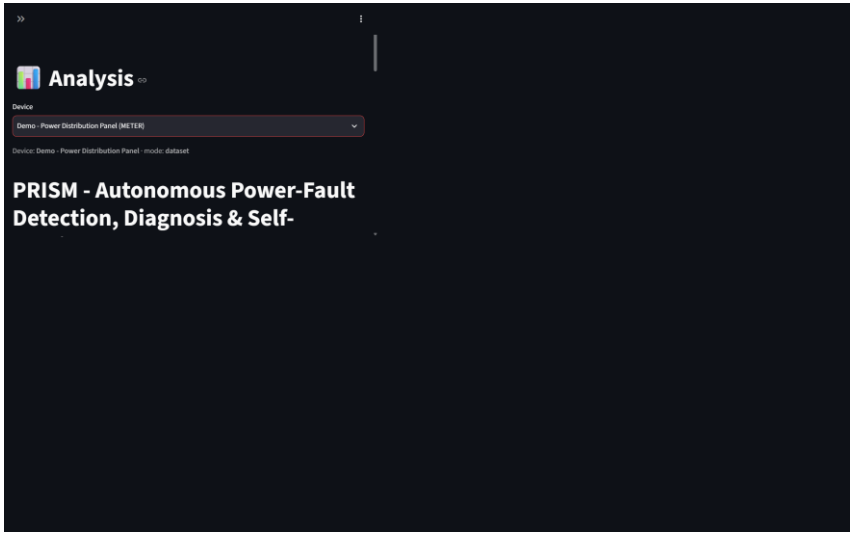
- Cloud Run - scale-to-zero; ~$0 when idle, pay only per request.
- Vertex AI (Gemini + embeddings) - per-token pricing; low for a demo workload.
- Firestore / Cloud Storage / BigQuery - minimal at demo scale (free-tier friendly).

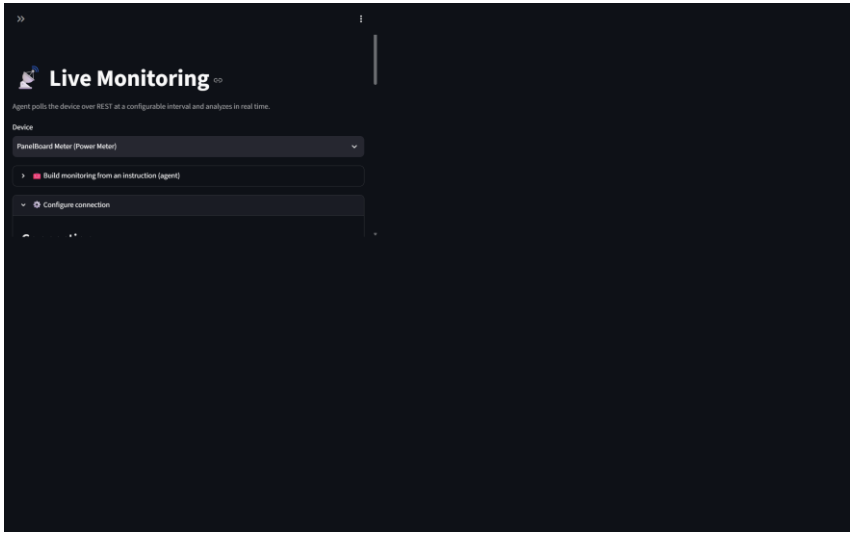
Estimated demo running cost: under $50 / month.

Production cost scales roughly linearly per device and per analysis, with no fixed server cost.

(Illustrative estimate - actual cost depends on usage, region and data volume.)

Snapshots of the prototype





Prototype Performance report/Benchmarking

- 121 automated tests passing (auth, RAG, generation, engine, live bridge, PRISM parity).
- Demo analysis: 3 assets with 3 injected faults -> 100% diagnosis accuracy vs ground truth.
- Illustrative impact: 223 minutes of downtime avoided and $153,550 cost avoided across 6 incidents, each scored against a no-platform baseline (synthetic scenarios, not realized savings).
- MTTR drops from tens of hours to 1-2 analysis cycles.
- Deployed and verified live on Cloud Run - Firestore-backed, Gemini on Vertex AI, zero runtime errors.

Additional Details/Future Development (if any)

- Real edge-sensor and IoT integration via Pub/Sub and Dataflow for always-on ingestion.
- Multi-device fleet modeling with topology correlation across an entire site.
- Predictive maintenance - forecasting failures before they occur.
- Support for additional industrial protocols (Modbus, DLMS and others).
- Mobile alerts and automated work-order dispatch; multi-tenant SaaS for operators.
- Domain-tuned models and on-device inference for low-connectivity sites.

Provide links to your:

GitHub Public Repository:

https://github.com/puneetpahuja-eaton/kairos-ai

Demo Video (3 minutes):

https://drive.google.com/drive/u/1/folders/1-W7LAChaeTS-c9UKNpMyIzHEPwEwOItQ

Final Product (deployed on Cloud Run):

https://kairos-ai-498775210896.asia-south1.run.app

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Thank you

